Supervyse OPF SNMP Developer's Guide

Revision 1.0

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2024-11-07	Dean Chien	Brad Chang	Initial version
1.1	2025-01-23	Dean Chien	Brad Chang	Initial version

1.SNMP Developer's Guide

1.1. Introduction

This guide is for users to realize how to customize SNMP implementation.

Create User's MIB File

- First, customers should have their own Private Enterprise Numbers (PENs). If not, they should apply to IANA for one by this hyperlink.

http://pen.iana.org/pen/PenApplication.page

- PEN is created and maintained by the Internet Assigned Number Authority (IANA) in a public registry.
- Enterprise numbers are commonly used in the Management information base (MIB) files of the Simple Network Management Protocol (SNMP) and in vendor suboptions of the Dynamic Host Configuration Protocol (DHCP).
- INSYDE PEN number is 45065. For other's overall Enterprise Numbers, please refer to this hyperlink.

http://www.iana.org/assignments/enterprise-numbers/enterprise-numbers

- Customers can refer to Insyde MIB file "BMC-MIB" to create their own MIB file.

* e.g. Below is the MIB file example for customers, and 98765 is the PEN number example.

```
$(OPF_SDK)/isc-src/snmp/example/EXAMPLE-MIB

  EXAMPLE-MIB

  EXAMPLE-MIB DEFINITIONS ::= BEGIN

  IMPORTS
      MODULE-IDENTITY, OBJECT-TYPE, enterprises, Integer32 FROM SNMPv2-SMI;

  test MODULE-IDENTITY
      LAST-UPDATED "201710031400Z"
      ORGANIZATION "www.example.com"
      CONTACT-INFO "email:example-mib@example.com"
      DESCRIPTION  "Example MIB objects for agent module example"
      ::= { enterprises 98765 }

  example    OBJECT IDENTIFIER ::= { test 1 }
```

1.2. Develop SNMP

The source code of SNMP is under $(OPF_SDK)/isc-src/snmp.

```
$(OPF_SDK)/isc-src/snmp
```

1.2.1. Example code

There is sample code for snmp get/set handler template.
User could reference for further SNMP OID creation or handler implementation.
The sample code is located in the "example" directory of snmp.

```
$(OPF_SDK)/isc-src/snmp/example/
```

1.2.1.1. SNMP Get

1. Add "example_get" description in MIB file.
   $(OPF_SDK)/isc-src/snmp/example/EXAMPLE-MIB, add example "example_get".

```
example    OBJECT IDENTIFIER ::= { test 1 }
...
--
--  MIB Module Object Types
--
example_get       OBJECT-TYPE
    SYNTAX        DisplayString
    MAX-ACCESS    read-only
    STATUS        current
    DESCRIPTION   "MIB example get"
    ::= { example 1 }
```

2. Define the OID for example_get and register the OID and handler with SNMP.
   $(OPF_SDK)/isc-src/snmp/ipmiAgentPluginObject.c, and add source code, custom PEN number, oid.
   a) Function netsnmp_register_instance() : register the OID and handler function.

```
oid snmpGetExample_oid[] = {1, 3, 6, 1, 4, 1, 98765, 1, 1, 0};
                                                     ^  ^
                                                     |  |----- example_get
                                                     |-------- example

void init_ipmiAgentPluginObject(void) {
   ...

netsnmp_register_instance(netsnmp_create_handler_registration("SnmpExampleGetHandler",
SnmpExampleGetHandler,
                                                                snmpGetExample_oid,
                                                          OID_LENGTH(snmpGetExample_oid),
HANDLER_CAN_RWRITE));
}
```

3. Define the handler function for example_get in header file.
   $(OPF_SDK)/isc-src/snmp/example/snmpHandlerExample.h, and add source codes.

```
int SnmpExampleGetHandler(netsnmp_mib_handler *handler,
                          netsnmp_handler_registration *reginfo,
                          netsnmp_agent_request_info *reqinfo,
                          netsnmp_request_info *requests);
```

4. Implement the handler function for example_get in c file.
   $(OPF_SDK)/isc-src/snmp/example/snmpHandlerExample.c, and add source codes.
   a) reginfo-mode
      The sequence stages to process SNMP get, usually we only implement cases of MODE_GET.
      i. MODE_GET
         In this case is to execute the real get action.

```
#include "snmpHandlerExample.h"

/**
    @brief SnmpExampleGetHandler

    snmp exmaple get handler

    @param[in] handler mib handler
```

```
    @param[in] reginfo hanlder registration
    @param[in] reqinfo agent request information
    @param[in] requests information request
    @return 0 for success, others for error.
*/
int SnmpExampleGetHandler(netsnmp_mib_handler *handler,
                          netsnmp_handler_registration *reginfo,
                          netsnmp_agent_request_info *reqinfo,
                          netsnmp_request_info *requests)
{
    netsnmp_request_info *request;

    for (request = requests; request; request = request->next) {
        if (request->processed)
            continue;

        switch (reqinfo->mode) {
            case MODE_GET:
                snmp_set_var_typed_value(
                    request->requestvb,
                    ASN_OCTET_STR,
                    "example_get",
                    strnlen_s("example_get", sizeof("example_get")));
                break;

            default:
                //DBG_LOG(LOG_INFO, "default\n");
                /* we should never get here, so this is a really bad error */
                break;
        }
    }
    return SNMP_ERR_NOERROR;
}
```

1.2.1.2. SNMP Set

1. Add "example_set" description in MIB file.
   $(OPF_SDK)/isc-src/snmp/example/EXAMPLE-MIB, add example "example_set".

```
example    OBJECT IDENTIFIER ::= { test 1 }
...
--
--  MIB Module Object Types
--
example_set       OBJECT-TYPE
    SYNTAX        Integer32
    MAX-ACCESS    read-write
    STATUS        current
    DESCRIPTION   "MIB example set"
    ::= { example 2 }
```

2. Define the OID for example_set and register the OID and handler with SNMP.
   Add source code and custom PEN number and oid in $(OPF_SDK)/isc-src/snmp/ipmiAgentPluginObject.c.

```
oid snmpSetExample_oid[] = {1, 3, 6, 1, 4, 1, 98765, 1, 2, 0};
                                                     ^  ^
                                                     |  |----- example_set
                                                     |-------- example

void init_ipmiAgentPluginObject(void) {
   ...
```

```
netsnmp_register_instance(netsnmp_create_handler_registration("SnmpExampleSetHandler",
SnmpExampleSetHandler,
                                                                    snmpSetExample_oid,
                                                              OID_LENGTH(snmpSetExample_oid),
                                                              HANDLER_CAN_RWRITE));
}
```

3. Define the handler function for example_set in header file.
   Add source codes in $(OPF_SDK)/isc-src/snmp/example/snmpHandlerExample.h.

```
int SnmpExampleSetHandler(netsnmp_mib_handler *handler,
                          netsnmp_handler_registration *reginfo,
                          netsnmp_agent_request_info *reqinfo,
                          netsnmp_request_info *requests);
}
```

4. Implement the handler function for example_set in c file.
   Add source codes in $(OPF_SDK)/isc-src/snmp/example/snmpHandlerExample.c.

   a) reginfo-mode
      The sequence stages to process SNMP set, usually we only implement cases of MODE_SET_RESERVE1 and MODE_SET_ACTION.
      i) MODE_SET_RESERVE1
         In this case is check data format is correct or not.
      ii) MODE_SET_ACTION
         In this case is to execute the real set action.

```
int SnmpExampleSetHandler(netsnmp_mib_handler *handler,
                          netsnmp_handler_registration *reginfo,
                          netsnmp_agent_request_info *reqinfo,
                          netsnmp_request_info *requests)
{
    netsnmp_request_info *request;
    int ret = 0;
    int set_value = 0;

    for (request = requests; request; request = request->next) {
        if (request->processed)
            continue;

        switch (reqinfo->mode) {
            case MODE_SET_RESERVE1:
                /*
                 *  Check that the value being set is acceptable
                 */
                DLOG_DEBUG("SnmpExampleSetHandler MODE_SET_RESERVE1\n");
                ret = netsnmp_check_vb_type(request->requestvb, ASN_INTEGER);

                if (ret != SNMP_ERR_NOERROR) {
                    netsnmp_set_request_error(reqinfo, request, ret);
                }
                break;

            case MODE_SET_RESERVE2:
                /*
                 *  This is conventially where any necesary
                 *   resources are allocated (e.g. calls to malloc)
                 */
                break;

            case MODE_SET_FREE:
                /*
                 *  This is where any of the above resources
```

```
                 *   are freed again (because one of the other
                 *   values being SET failed for some reason).
                 */
                break;

            case MODE_SET_ACTION:
                /*
                 *  Set the variable as requested.
                 *   Note that this may need to be reversed,
                 *   so save any information needed to do this.
                 */
                DLOG_DEBUG("SnmpExampleSetHandler MODE_SET_ACTION\n");
                set_value = *request->requestvb->val.integer;
                DLOG_DEBUG("SnmpExampleSetHandler example set %d\n", set_value);
                break;

            case MODE_SET_UNDO:
                /*
                 *  Something failed, so re-set the
                 *   variable to its previous value
                 *  (and free any allocated resources).
                 */
                DLOG_DEBUG("SnmpExampleSetHandler MODE_SET_UNDO\n");
                break;

            case MODE_SET_COMMIT:
                /*
                 *  Everything worked, so we can discard any
                 *   saved information, and make the change
                 *   permanent (e.g. write to the config file).
                 */
                DLOG_DEBUG("SnmpExampleSetHandler MODE_SET_COMMIT\n");
                break;
            default:
                //DBG_LOG(LOG_INFO, "default\n");
                /* we should never get here, so this is a really bad error */
                break;
        }
    }
    return SNMP_ERR_NOERROR;
}
```

1.3. Example code verification

Launch OPF image and do verification with SNMP commands through Ubuntu client.

1.3.1. Example_get

Verify "example_get"(OID: 1.3.6.1.4.1.98765.1.1.0) functionality using snmpget command in Linux. Expect result is to return the string "example_get".

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(BMCIP)
1.3.6.1.4.1.98765.1.1.0
```

Output:

```
SNMPv2-SMI::enterprises.98765.1.1.0 = STRING: "example_get"
```

1.3.2. Example_set

Verify "example_set"(OID: 1.3.6.1.4.1.98765.1.2.0 i 1) functionality using snmpset command in linux. Expect result is to return the integer 1 that command just set.

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(BMCIP)
1.3.6.1.4.1.98765.1.2.0 i 1
```

Output:

```
SNMPv2-SMI::enterprises.98765.1.2.0 = INTEGER: 1
```